Kubernetes Operations Guide

Container Recovery

Kubernetes can restart failed or unhealthy containers. A Deployment uses ReplicaSets to maintain the desired number of Pods. If a Pod is removed or a container fails, the controller works to restore the desired state.

Health Checks

Liveness probes help Kubernetes determine whether a container should be restarted. Readiness probes indicate whether an application is ready to receive traffic. These mechanisms allow Kubernetes to detect unhealthy workloads and control traffic appropriately.

Scaling

Horizontal Pod Autoscaling can adjust the number of Pod replicas based on observed metrics. Scaling changes the desired number of replicas, while the Deployment controller ensures that the requested number of Pods is maintained.

Troubleshooting

Common troubleshooting steps include checking Pod status, reviewing container logs, inspecting events, checking Service endpoints, and validating resource requests and limits.